CATERPILLAR®

DEALER BEST PRACTICE SERIES

Repair Management

Application | **Maintenance** | **Site Management** | Component Rebuild | Component Life Management | Safety | MARC Management

June 2011
1007-2.0-1107

1.0 Introduction

This Best Practice is part of a series of Strategic Best Practices that are directed to address in detail the ten (10) different processes of the Caterpillar Global Mining Maintenance & Repair Process recommended to manage your on site mining operations.

Repair management is the functional element that is dedicated to the execution of all the maintenance activities on the equipment. These activities can be planned or unscheduled, requiring from the on site organization the appropriate preparation in terms of resources and skills to respond adequately.

"Adequate Resources Organized & Managed to Perform Efficient & Effective Repairs"

There are two distinct service areas that Repair Management must consider in a mining operation, the shop and the field. Because of the nature of the services performed in these two areas, effective mines recognize the need of organizing the support in the same way, developing & assigning specialized crew and equipment to each of them.

Field Service Operation

Shop Operation

Adequate human resources (HR), both in quantity and quality (appropriate skills), organized effectively (organization structure), and equipped with the right tools and facilities are the critical contributing areas to the performance results of your repair management function.

The Right People, well equipped supported by adequate Facilities… key for Success in Repair Management.
Photo: Courtesy of Sotreq, Brazil – Samarco On Site Crew

2.0 Best Practice Description

The scope and characteristics of the Repair Management (RM) function will be determined by the maintenance activities / routines that are going to be performed in a particular mine site. The first step in designing and implementing the Repair Management function is to analyze the Maintenance Strategy to determine the characteristics of the service required.

Mine Production Requirements

CSA's definitions

Fleet Characteristics

Site Characteristics

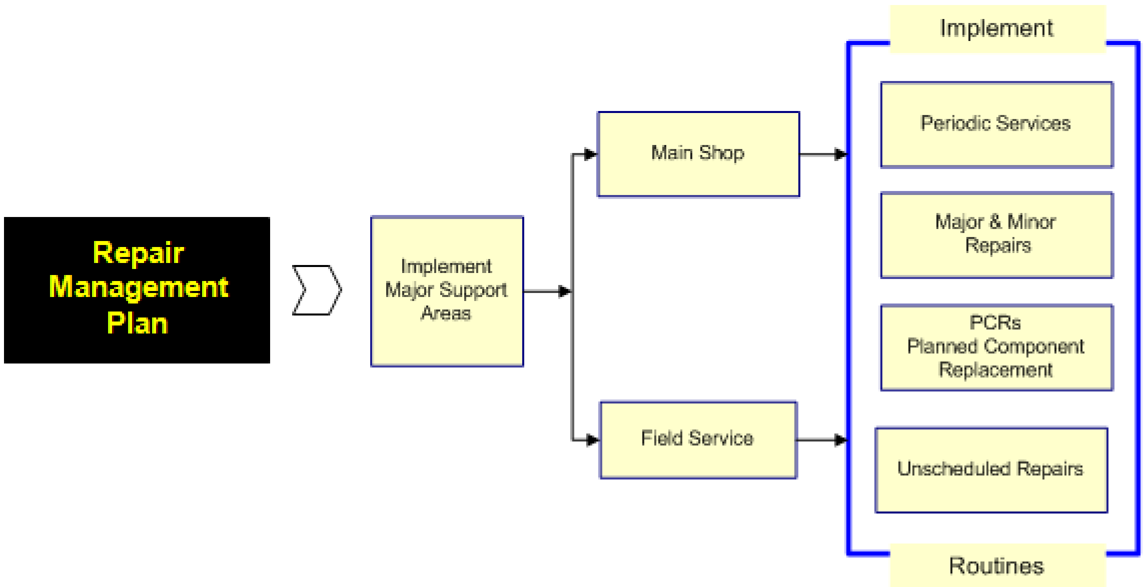
External Support

Maintenance Strategy

Establish the Repair Management Plan

(What is what we have to accomplish and How…)

In the Repair Management Plan, we outline all the necessary procedures and resources to perform the Maintenance & Repair activities efficiently and effectively. - "What are we going to do? How are we going to do it? What resources will we need?"



Mining mobile equipment are subjected to maintenance and repair activities that can be executed in the Field or in the Shop. Our strategy has to consider these two main areas, Field and Shop. The activities that normally take place in these two areas are:

Field Service

- Diagnostic of unscheduled machine stops
- Short / small repairs
- Repairs that don't require the opening of a component or system
- Visual observation of machine application and operation
- Coordination of all Field machine related activities

Shop Service

- Scheduled machine maintenance & repair, including
    - Preventive Maintenance
    - Component Exchange (PCR)
    - Major & Minor Repairs (Backlogs executed outside the normal windows of opportunities)
- Unscheduled Repairs
- General coordination of the entire maintenance operation

Field Service

The Field Service is a very dynamic function that must respond quickly and effectively to all types of service calls. The right personnel properly equipped and trained in the correct working processes are key issues to consider at the time of the implementation of the Field Service.

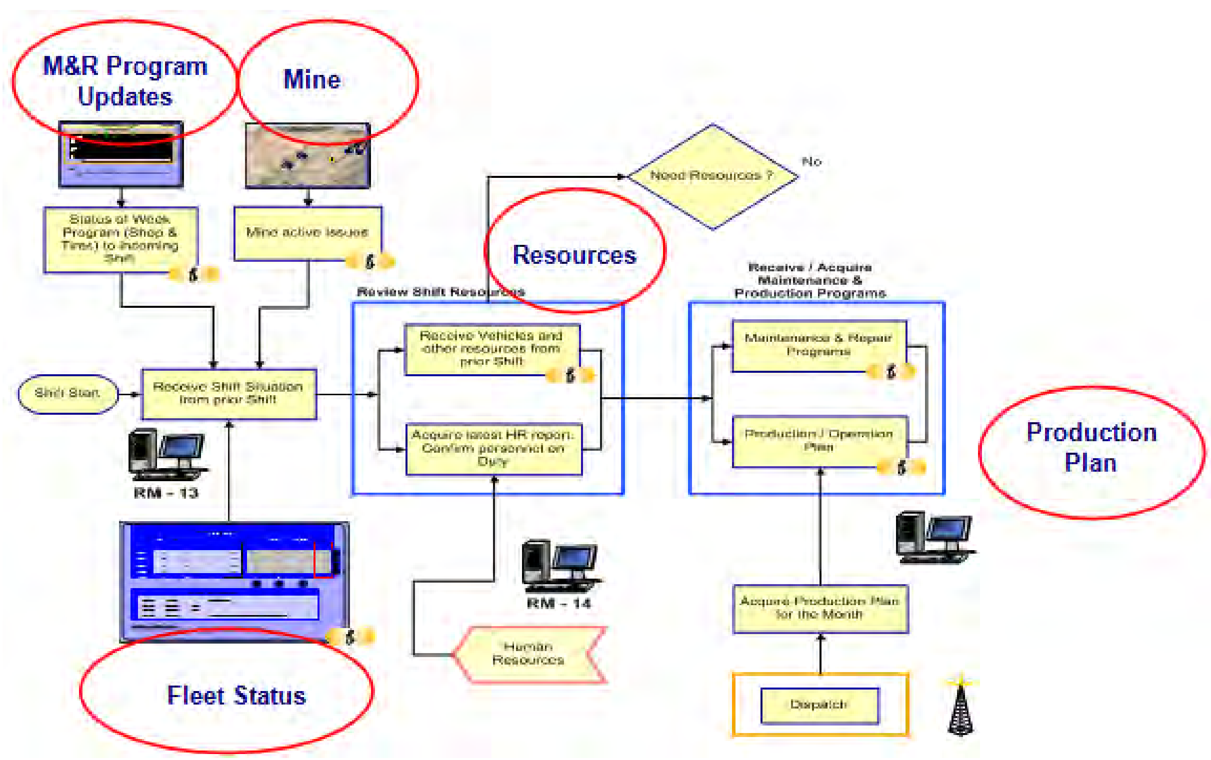
The first process to review and understand is the overall Field Shift operation, the tasks that the field crew performs to keep repairs progressing from one shift to the next. Many sites overlook the importance of an effective handoff of work. Preparing the work and information to ensure the continuation of the field support in the upcoming shifts is often as important as the work actually being done on a given shift. As there are multiple shifts to cover the 24 / 7 mine operation, effective handoff of work is vital to an efficient Field Service operation.

An efficient and effective shift passing is a must and is based on accurate information. We recognize three groups of information that must be present in the shift passing process:

1.- Equipment Related
- Fleet Status Report
- Updated Maintenance & Repair Program (Week)

2.- Mine / Production Related
- Mine active issues (Application and/or Operation)
- Production / Operation Plan
- Mine accessibility

3.- Resources
- Personnel



1.- Equipment Related Information

The incoming shift must quickly identify the status of all the equipment in the fleet and the Field Service priorities to follow. The Equipment Status Report is a commonly used tool for this purpose.

Mine Operation ABC
Fleet Status

Shift: A Date: 18-Sep-06

	Total Equip.	Operating	Available	Avail %	Down	Shop	Tires	Field
Fleet 1	17	13	1	82%	3	2	1	0
Fleet 2	14	11	1	86%	2	1	0	1
Fleet 3	9	9	0	100%	0	0	0	1
Fleet 4	17	15	0	88%	2	1	0	1
Total	57	48	2	88%	7	4	1	3

Equipment operating, on stand by, or down and location. It also provides a balancing count to ensure all equipment are considered / included.

Service Field Vehicles

Vehicle	Status	Observations
FS - 101	Available	OK
FS - 102	Available	OK
FS - 103	Available	Scheduled for PM; Replacement already requested.

A quick summary of the support equipment status is also included in the report.

General Observations:

- Priority 1 is Truck 107. Located in Blasting Area.
- Truck 287 scheduled for 200 h PM at 5:00 AM

Service priorities

Example of a Field Service Report

The Fleet Status Report gives a quick and effective view of the current status of all the equipment of the operation. Now the M&R Schedule for the week gives the other portion of the information needed. "What equipment is scheduled to be down?"

Maintenance & Repair Program Week: 43

Date: September, xx, 2006

Time	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday
05:00	110 - PM2	108 - PM[illegible]	103 - PM1	106 - PM1	101 - PM2		
07:00							
09:00			109 - PM1	108 PCR	108 PCR	108 PCR	108 PCR
11:00		107 - PM3					
13:00	105 SR						
15:00							
17:00							
19:00							
21:00							
23:00							
Downtime - Availability Projection							
PM h	8	16	16	8	8	0	0
Backlogs	2	4	8	0	0	0	0
# Backlogs							
Scheduled h	8			20	24	24	24
Unscheduled h	12	12	12	12	12	12	12
Total h	30	32	36	40	44	36	36
Est. Availability	88%	87%	85%	83%	82%	85%	85%
Labor Hours Required - Projection							
MH - PMs	40	80	80	40	40	0	0
MH - Backlogs	12	24	32	8	8	0	0
Scheduled h	24	0	0	60	72	72	72
Unscheduled h	48	48	48	48	48	48	48
MH - Required	124	152	160	156	168	120	120
MH - Available	150	150	150	150	150	150	150

Notes:

- Truck 105 is still due for this week. We will confirm the new date for entering this unit to the Shop.

The updated M&R program is submitted daily to the Field Service crew.

Field Service follows up the program and assists with the stoppage of the machines when needed.

2.- Mine / Production Related Information

The second type of information the Field Service crew needs is that related to the mine operation. Where are the machines working? What is the safest / quickest access to the working areas? Are there any safety issues? Are there any designated blasting locations or times? Etc.

We recommend considering three types of mine related information:

- Production / Operation Plan. The production plan identifies the current and future mine work zones.

- Mine accessibility. The access roads in open pits mines are very dynamic, changing constantly according to the production needs. The Field Service crew has to be updated regularly. A good practice is to include this review in every shift changes.

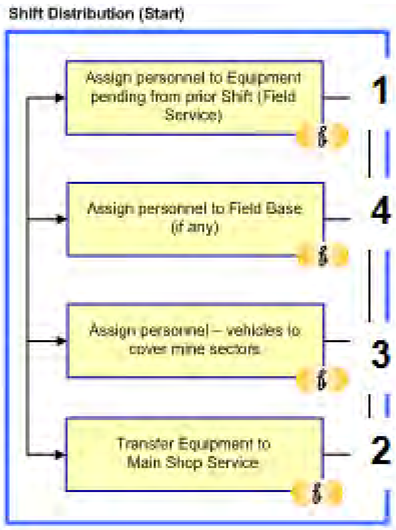
- Mine active issues (application and/or operation). Applications and operation related issues that were detected through direct observations or condition monitoring analysis and are currently active.

3.- Resources.

What are the resources the shift has to run the shift (personnel and equipment). The resources situation has to be defined quickly to correct any shortcomings with the central maintenance shop. The shift has to be fully equipped in a short time, to be ready to continue the service of the work in progress and accept new calls.

Good example of a well-equipped field vehicle.

All necessary information is received, the resources reviewed and augmented as necessary. Now is time to organize and distribute the work,



1 - 2 Based on the equipment that are down, two first actions are taken. Personnel is assigned to specific machines being serviced in the field and if necessary some services are transferred to the main shop.

3 - 4 The rest of the Field Service crew is organized to cover the different sectors of the mine and the field base if one is established.

The mine sectors are assigned to different Field Service crew teams. This is just the starting point. During the shift operation the service – team coordination is constant and very dynamic. Communication, within FS teams and the dispatch is very critical.

In some operations and because of the extension of the mine, a good idea is to establish a field base, dedicated to short services and to reduce the pressure on the mobile field service teams.

The oncoming shift receives the information from the former shift. The resources are reviewed and augmented if necessary and the personnel are organized to cover all production areas of the mine. Now it's time to run the shift, to perform the tasks that normally takes place during the shift.

Receive new service Requests	The first responsibility is to accept and provide support to new calls. These calls are unscheduled and the attention priority is given by operations. The Field Service crew accepts the call, records it in the shift log and assigns personnel according to resource availability and location.
Control execution of Field Work	The work in progress in Field Service can be happening in multiple locations at the same time. An efficient follow up of all the work in progress must be performed. Critical points to observe are the diagnostic /determination of the problem and needed repairs and the decision to repair the machine in the Field or send it to the Shop. The Field as we said before is the first point of contact from maintenance and is dedicated to small / short repairs that don't require extensive labor or disassembly of components / systems of the machine.
Coordinate scheduled equipment to main shop	The Field Service is the "owner" of all equipment in the Field, maintenance wise. FS has to be informed of all equipment scheduled to the shop or other repair centers, so they know of the destination and reason of the repairs and can assist in the stop or movement of the machines if necessary.
Supervise / control equipment to Tire Service	In the same way the Tire Shop program (Shop & Field) has to be communicated to Field Service for the same reasons as above.
Perform application/operation observations	FS crew is constantly in the mine driving all existing roads. We encourage incorporating in the FS responsibilities the visual detection of the mine application and operation issues that can be pursued to eliminate or reduce the impact on the equipment.
Supervise / support other Field Maintenance related activities	As we stated before, Field Service is the "owner" of all maintenance activities that happen in the field. Field Service has to be informed and their responsibilities includes the follow up and general supervision of the execution of the work scheduled.

Shop Service

The shop is where the majority of the maintenance activities are executed and coordinated. These maintenance activities can be scheduled and / or unscheduled and include:

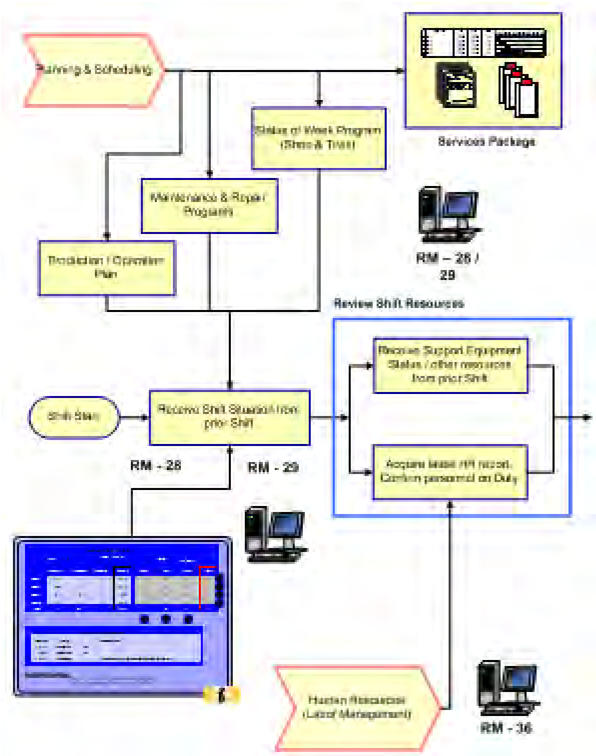
In the case of scheduled activities:

- Preventive Maintenance
- Component Exchange (PCR)
- Major & Minor Repairs

And for the unscheduled activities:

- Major, long repairs that cannot be handled in the field
- Any kind of repairs that represent a risk of contamination of any system or compartment on the machines
- Any repair that because of the work load of the field crew needs to be taken under the responsibility of the main shop

Fleet characteristics, facilities, logistics, labor availability, level of competencies, site characteristics, etc. are the type of inputs that are captured in the maintenance strategy and are used to define and organize the work force and resources in the main shop. These resources are normally organized and distributed in multiple shifts to properly support the mine operation, that normally works under a 24 hours 7 days a week regimen.



It is a continuous operation of multiple shifts that has to be managed correctly to ensure efficiency at all times. Shift by shift, the work on the equipment is completed. In the same way as in the field service operation discussed previously, efficient shift passing is a must.

In the case of the shop, the key information that is required at the beginning of each shift is similar as that identified in the field service, but with slightly different objectives:

1.- Machine related Information

- Fleet Status Report
- Work in progress, work performed and pending (shift report).
- Updated Maintenance & Repair Program
- Service Packages for machines that are scheduled for service.

2.- Mine / Production Related

- Mine active issues (Application &/or Operation)
- Production / Operation Plan
- Mine accessibility

3.- Resources
- Personnel
- Facilities
- Support Equipment

As stated before, the information the shop operation needs for the shift start up is very similar to that analyzed in the field service section, but because of the nature of the work that is performed in the shop, this information must be focused in a different way to cover the continuation of longer repairs and the scheduled work.

1.- Machine Related Information

In this case the shop must always have a clear picture of what is the status of all machines and any work that is being executed in the shop and all other related areas, such as the tire shop, welding, the field, etc.

There are two main means that shift operations maintain this information: Dedicated Bulletin Boards and the Shift Report. Both compliment each other.

This is a good example of a Fleet Status Board currently in use in Samarco, Sotreq operation in Brazil.

While the status board provides a quick view of the status of all machines in the fleet, the shift report is used to summarize the work already completed and any remaining tasks that are necessary to complete the maintenance or repair.

The other pieces of information that are also included at this point are the updated Maintenance & Repair Program and the corresponding Service Packages. The M&R program contains the list of machines with the services scheduled and the service packages include all the needed instructions to perform those scheduled services or repairs.

2.- Mine / Production Related Information

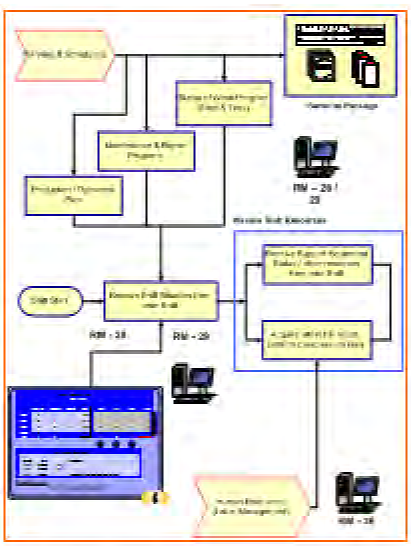
In the case of the shop, this information is made available as a reference to be used if necessary. Normally when the shop personnel have to perform any work in the field they are assisted by the field service personnel.

3.- Resources

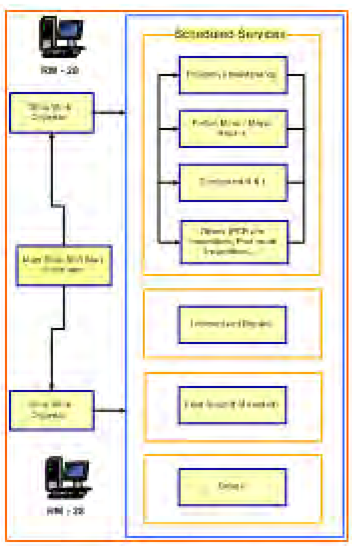
The information related to resources is much more critical in the case of the shop. The shop operation involves the use of multiple resources, such as facilities, special equipment, tooling, and of course the personnel scheduled and available to perform the work.

Every shift has distinct steps or phases that are important to observe. We recognize 8 different stages during the shift period:

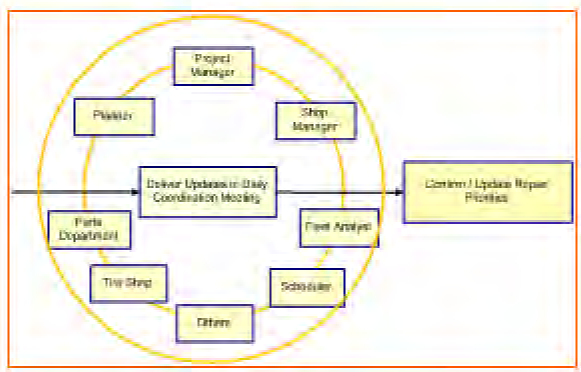
- Shift Passing. The incoming shift receives the information from the outgoing shift.
- Shift Organization. The resources are organized based on the jobs to be performed.
- Communication. The news and general maintenance operation status is communicated in the daily coordination meetings.
- Setting Priorities. Review / Define / confirm priorities for work continuation.
- Shift Operation. Executing the services and repairs as planned.
- Complete Records. Close work orders and complete necessary records.
- Next Shift Preparation. Prepare information for incoming shift.
- Shift Passing. The incoming shift receives the information of the work status and priorities.



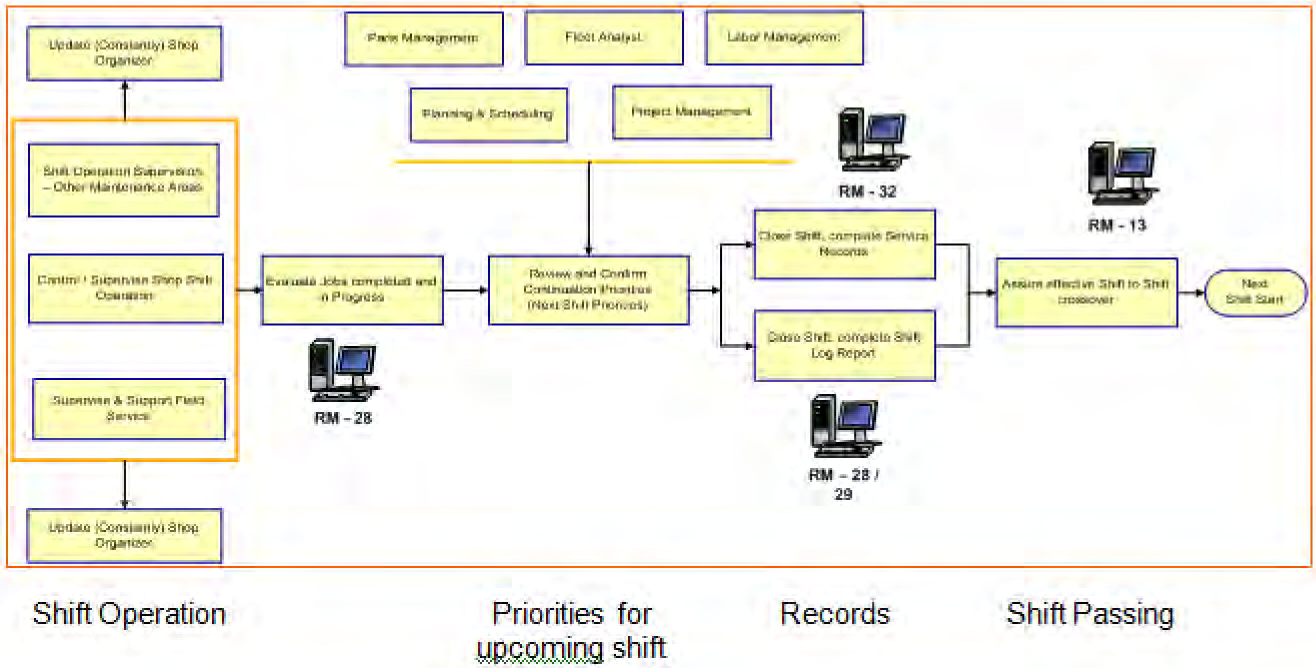
Shift Passing



Shift Organization



Communication and Priority Setting

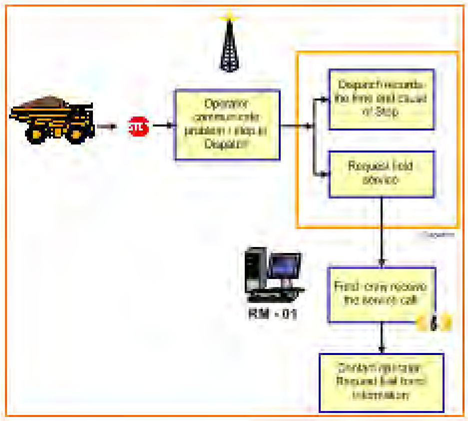


Now that we have the shifts organized and functioning in both operational areas (field and shop), it is time to look at two typical types of services that take place in these areas … the scheduled and / or unscheduled repairs.

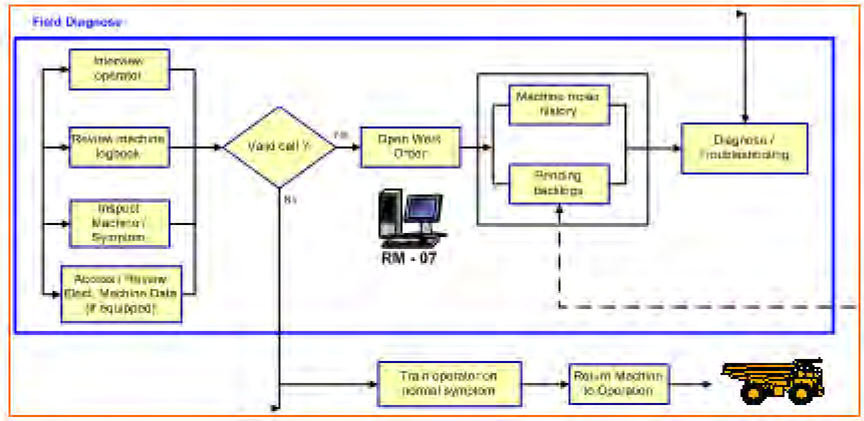
Unscheduled Repairs

An unscheduled stop by definition is an event that happens at random. The chain of events or stages that are important to consider in this type of service are:

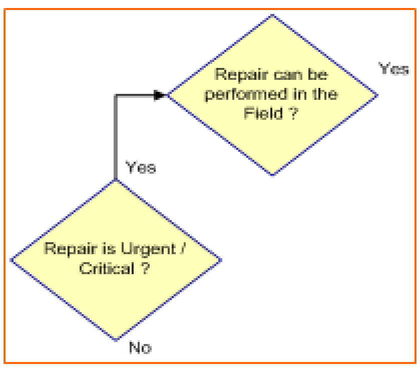
- Operator communicates machine down.
- Dispatch requests service to the FS crew.
- Initial assessment of the problem and call validation.
- Troubleshooting of failure.
- Decision: Repairs must be done immediately and can it be done in the field or shop?
- Based on the result of the assessment three actions can be taken:
    - Repair in the Field.
    - Generate a Backlog.
    - Transfer machine and repair to the Shop.
- Generate records.
- If repair is completed in the field, the machine is release to operation.



Machine Down, Operator Communicates to Dispatch, Dispatch requests service

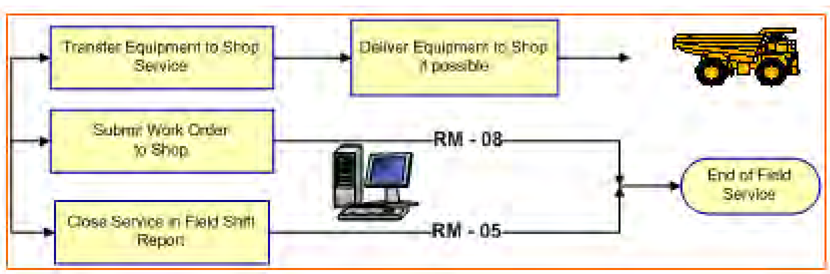


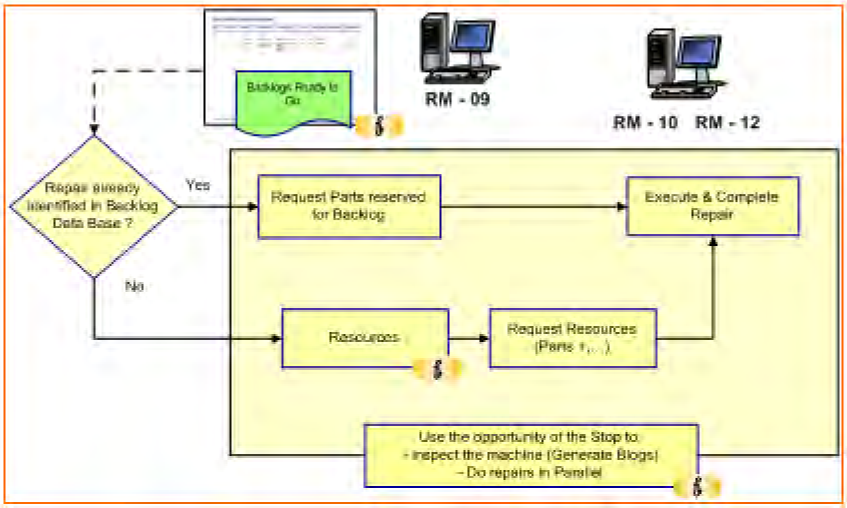
Initial assessment and problem validation, Troubleshooting / diagnosis of failure



The decision of making the repair immediately and performing it in the field or the shop has to be made as soon as the diagnosis defines the repair actions needed.

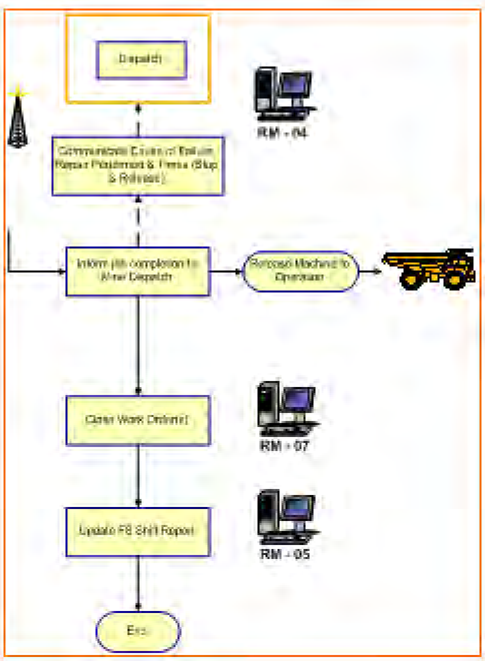
If the decision is to send the repair to the shop, the equipment with the supporting documentation (Work Order) must be transferred. In some cases, the machine can be moved to the shop in some others the machine remains in the mine.





At the time of executing the repairs, the Backlogs Pending and Ready to Go lists must be reviewed to see if the repair was already covered by a Backlog.

The Backlogs Ready to Go list must be reviewed to see if there are any Backlogs that can be done in parallel during the window of opportunity of the unscheduled stop.



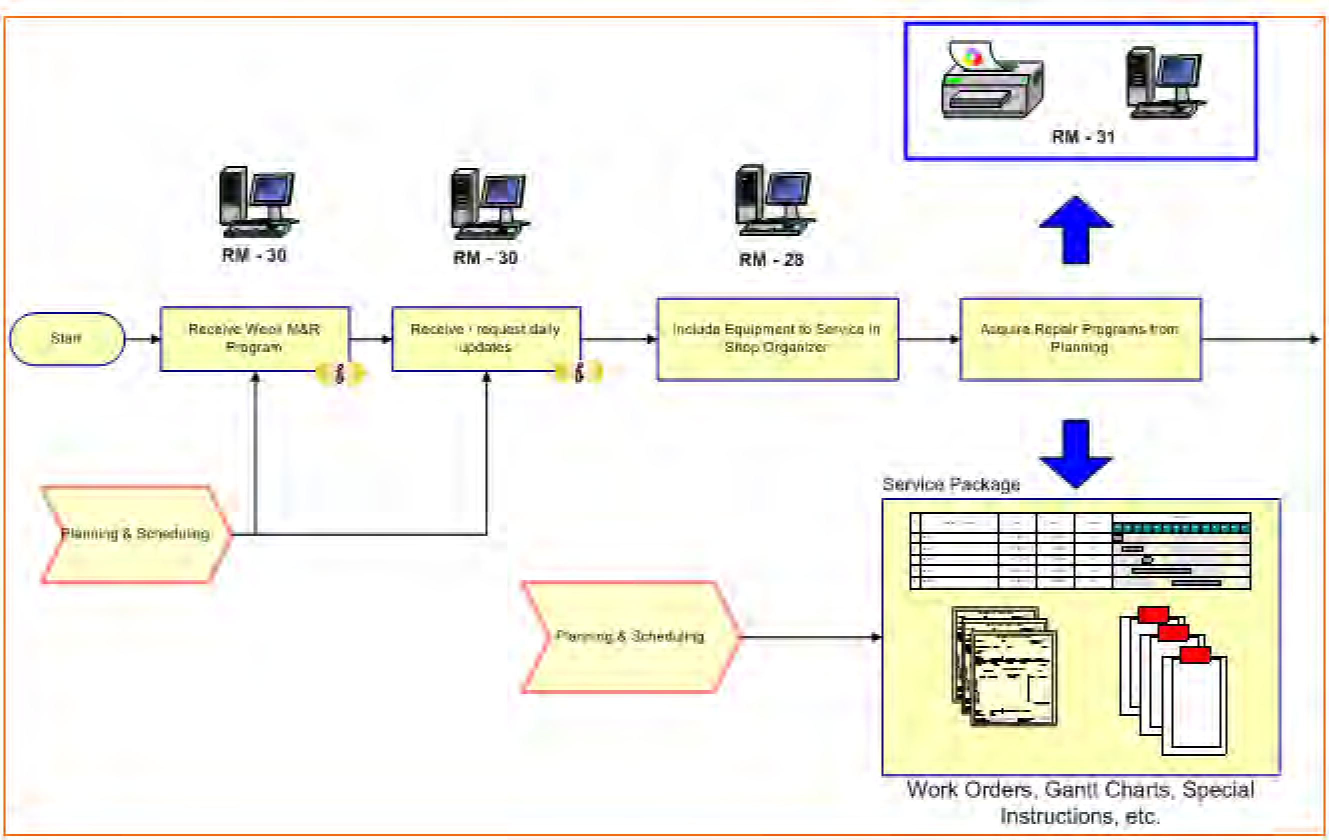
At the end of the repair(s), the machine is released to operation, the mine dispatch is informed and all maintenance records are completed and closed for that particular machine. The records to be completed are normally the work order and the shift reports (field and shop if it is the case).

Scheduled Repairs

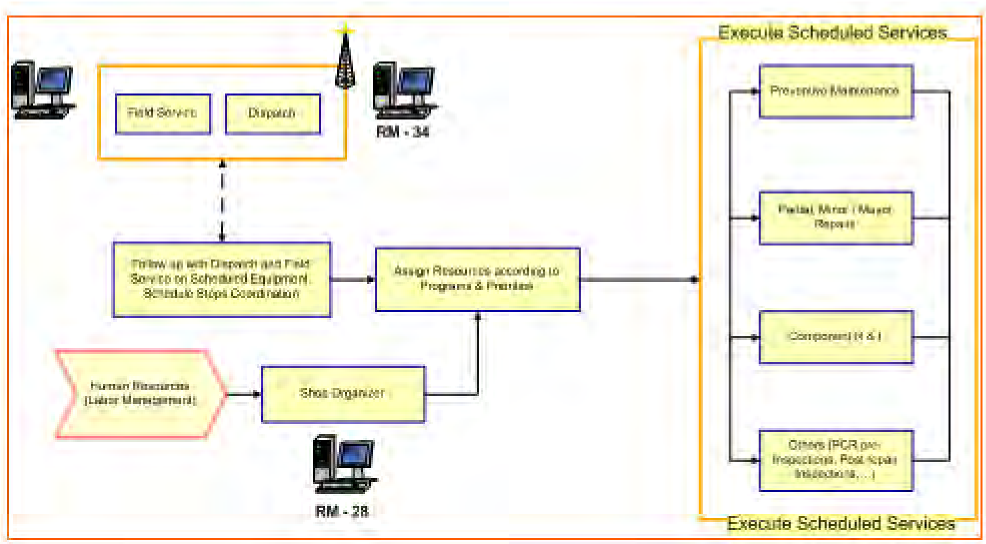
In the case of Scheduled repairs, this type of service happens in the Shop and involves the following chain of events:

- Machine / Service is included in the M&R program or is requested in advance.
- Machine stop and transfer to the shop is coordinated with Dispatch.
- The scheduled repairs / services are executed according to the plan.
- Pending tasks are passed to the next shift if necessary.
- Repairs are completed and a post repair inspection is peformed.
- Close documentation (WO, checklists, Blogs, shift reports…)

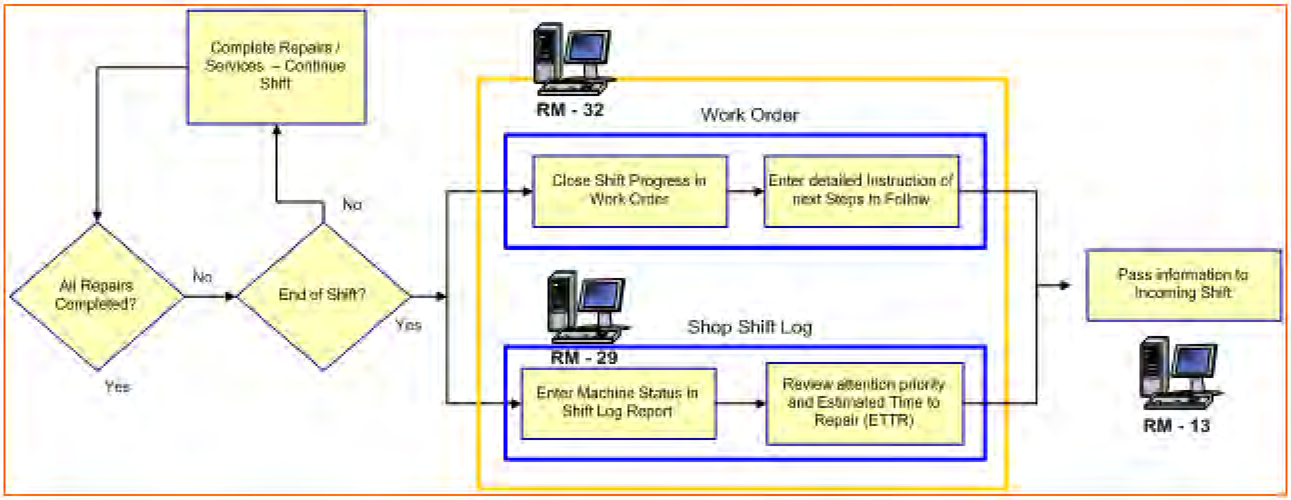
- Release machine to work.



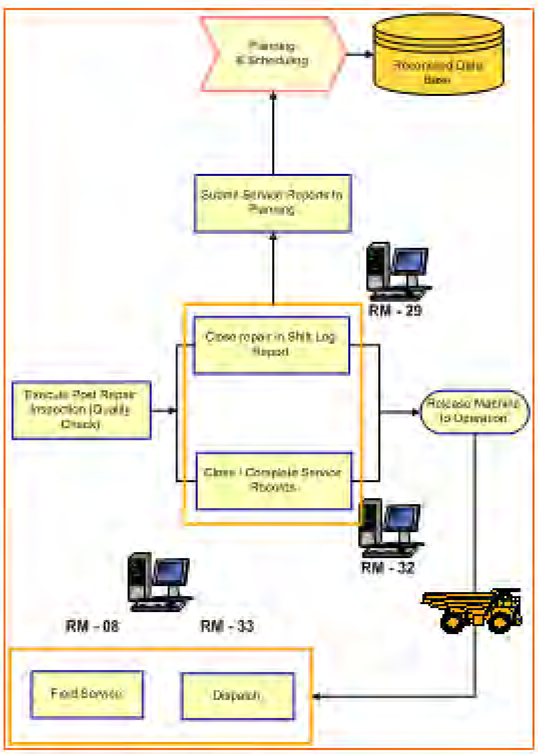
In this case, the repair or service is included in the weekly M&R program and the repairs to be performed are precisely defined in the dedicated documentation for a specific machine. Work Orders, Gantt Charts, Special Instructions, etc., are included in what we call the “Service Package”.



The equipment is released to the shop and the scheduled services are executed.



In the case of repairs that are scheduled or last for more than a shift, the information has to be passed from shift to shift until the tasks are completed. The outgoing shift reports on what was performed and more importantly on what are the next steps that the incoming shift has to continue working.



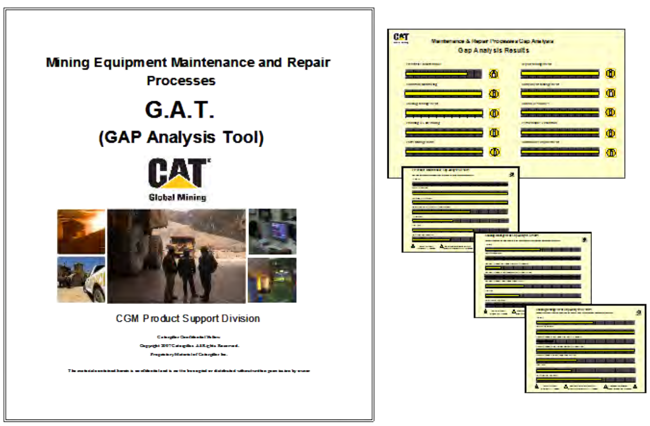
At the completion of the repairs, the shop crew must test the machine and perform the post – repairs inspections. The machine is then released to operation. The mine dispatch is informed and the maintenance records are closed and submitted at the end of the shift to Planning.

3.0 Implementation Steps

The implementation steps to follow will depend on the current situation of your mine site operation. What we are recommending as a first step is a thorough evaluation of your maintenance operation. Determine the “As Is” situation. Compare with the recommended model. Determine areas of opportunities and finally establish an action plan to correct or improve those areas of opportunities.

1.- Evaluate Your On Site Maintenance Operation

- Apply GAT (Gap Analysis Tool) - available through Global Mining Product Support division



The GAT is a survey dedicated to evaluating the 10 different processes recommended by Caterpillar Global Mining.

2.- Establish the “As Is” Situation

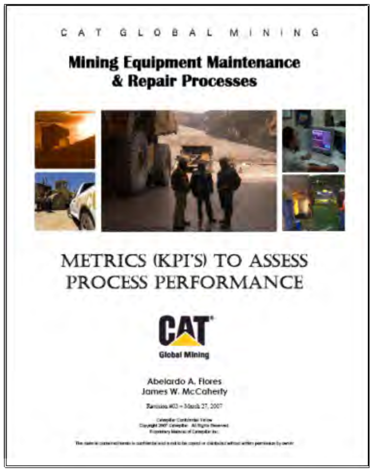
In establishing the “As Is” situation, we recommend performing two distinct activities:

- Determine the Process Flow Diagrams of all your processes implemented on site.

Based on the observations guided by the GAT tool and the on site investigation, draw and document the current processes on site.

- Establish the Baseline.

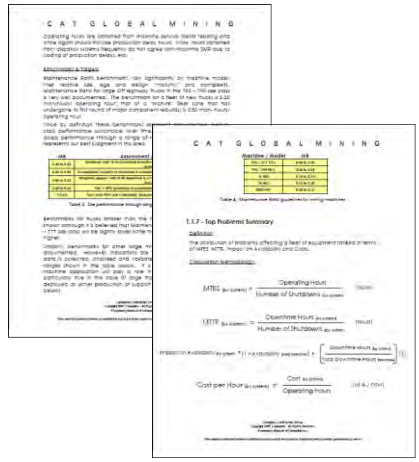
Measure the current performance results and document them for future reference. Apply the set of metrics recommended in CGM document dedicated to “KPI’s for Evaluating Maintenance and Repair Processes”


C A T G L O B A L M I N I N G

Mining Equipment Maintenance & Repair Processes

METRICS (KPI'S) TO ASSESS PROCESS PERFORMANCE

CAT Global Mining



This document contains the KPI's included in the GAT, calculation methodology and interpretation guidelines.

3.- Determine the Areas of Opportunities

Determine the areas of opportunities by comparing the "As Is" situation with the recommended processes detailed in this Strategic Best Practices and the Process Maps available through Cat Global Mining.

4.- Define the Action Plan

- Document your findings. The use of a 3-W form is recommended to record your findings, opportunities, and plans. The 3-W form will define, "What needs to be done", "Who will do it", and "When it will be completed".
- Devise and document the solutions for improving or correcting the weaknesses. Use the related BP's to support your search for ideas / solutions. Use a separate column in your 3W form to record the solution or actions to be taken.
- Analyze and determine resources needed, time to implement, and expected benefits.
- Prioritize the implementation of the different actions based on Return On Investment (ROI) results.
- Assign responsibilities and expected completion dates for each of the activities.

5.- Get the "Buy-In" of the On Site Organization

It is very important to get the buy-in of the on site organization. They are the ultimate "process owners" of the maintenance & repair strategy. Present the action plan, the implementation strategy and the expected benefits. Involve them in the implementation of the solutions and plan an in-depth training of all the involved personnel.

6.- Implement

Execute the implementation plan.

7.- Follow up

Follow up closely the implementation of your action plan.

- Set up “check points”.
- Review compliance of committed dates for tasks completion.
- Receive & analyze work progress from the person specifically responsible.
- Reinforce the positive results.
- Identify and eliminate threats and obstacles.
- Support the ownership of the on site organization
- Support the positive change of personnel. Correct the negative attitudes or opposition.
- Make adjustments as necessary.

8.- Measure the Results – Improve / Adjust If Needed

It is absolutely necessary to measure the results of the improvements implemented at the level of each individual process and at the final outcome, fleet performance.

Measure and trend the behavior of the KPI’s selected for establishing the “baseline” when the “As Is” situation was established. Compare and adjust as necessary.

4.0 Benefits

Repair Management is the function in the overall Maintenance & Repair process dedicated to the execution of repairs. It is where all the plans and strategies are executed. A properly structured organization must be adequately equipped with tooling, equipment and appropriate facilities. It must follow clear and effective working procedures. It is the foundation necessary to meet the goals of efficiency and effectiveness. The BP presented here and the detail documentation of the Repair Management flow diagrams will give you the information to size the task before you.

Repair Management is another important part of the puzzle of the recommended Maintenance and Repair Processes supporting the ultimate goal of *safe and productive operation of the mining equipment at the lowest possible cost per ton.*

5.0 Resources Required

The resources required will vary widely with the characteristics of your operation, such as size of the fleet of equipment, geographical location, complexity, maintenance strategy, etc. All these factors will in the end determine the type of organization and the facilities and equipment that will be needed.

In terms of human resources, there are some key positions that must be considered at the time of defining and structuring the organization.

M&R Operations Supervisor	***Leads the M&R operation to assure the effective and efficient support of scheduled and unscheduled repairs***
Shop Shift Supervisor	***Coordinates and directs shop resource priorities to meet daily repair goals***
Shop Technician	***Executes shop repairs***
Field Shift Supervisor	***Coordinates and directs field maintenance and repair resources… Boss of the field operations during his shift.***
Field Technician	***Performs field diagnosis and repairs***
PM Supervisor	***Manages daily preventive maintenance plan***

6.0 Supporting Attachments / References / Resources

The following documents are available to support the use of KPI's when measuring performance results of mining equipment and M&R processes.

Performance Metrics for Mobile Mining Equipment

Metrics (KPI's) to assess process performance

7.0 Related Best Practices

Cat Global Mining M&R Processes	BP **1007-2.0-1100**
Preventive Maintenance	BP **1007-2.0-1101**
Condition Monitoring	BP **1007-2.0-1102**
Backlog Management	BP **1007-2.0-1103**
Planning & Scheduling	BP **1007-2.0-1104**
Parts Management	BP **1007-2.0-1105**
Component Management	BP **1007-2.0-1106**
Human Resources / Training	BP **1007-2.0-1108**
Performance Evaluation	BP **1007-2.0-1109**
Continuous Improvement	BP **1007-2.0-1110**

8.0 Acknowledgements

This best practice was written by:

Abelardo A. Flores
Sr. Equipment Management Consultant
Caterpillar Global Mining
Flores_Abelardo@cat.com
Phone: 1-309-360-2767